

Probabilistic Graphical Models

CSE 5830

Representation

Lecture 4

Local probabilistic Models (CPD)

Quiz 1 – Fill out form

Quiz 1

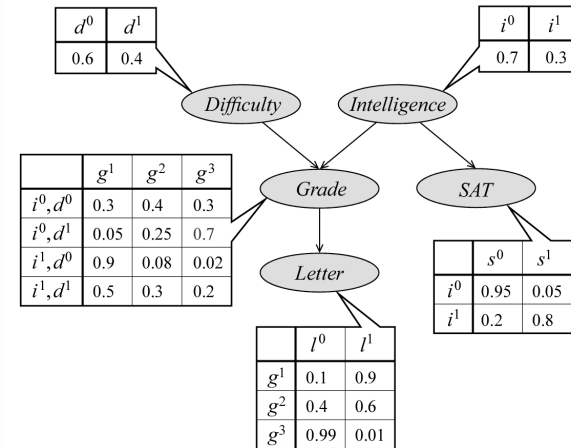


1-In the Bayesian Network shown below, D, I, S, and L are binary variables, and G has three possible values.

i-How many parameters we need using the conditional parameterization?

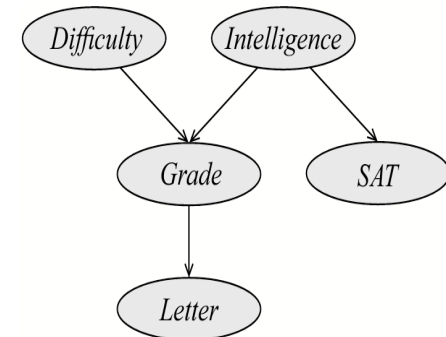
ii-How many parameters we need using the joint probability?

- a) i: 8, ii: 15
- b) i: 15, ii: 48
- c) i: 15, ii: 32
- d) i: 32, ii: 48



2-Which of the following independency is not accurate if Grade is given?

- a) $L \perp D | G$
- b) $L \perp S | G$
- c) $D \perp S | G$
- d) $L \perp I | G$



Goals for Today

- Conditional Probability Distributions (CPDs)
 - Structure in CPDs
 - Deterministic CPDs
 - Tree-structured CPDs
 - Noisy OR / AND
 - Linear Gaussians & generalizations

Local independencies and BN (review)

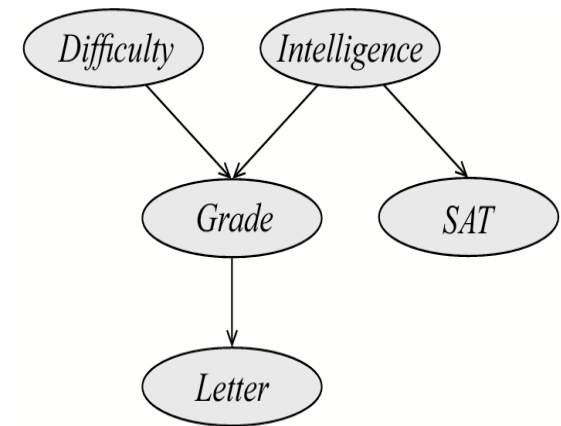
- A BN structure G encodes local independencies
- For each random variable X_i :
 $(X_i \perp \text{NonDescendants}(X_i) \mid \text{Parents}(X_i))$

- Joint distribution factorizes over G

$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i \mid \mathbf{Parents}(X_i))$$

Conditional Probability Distributions

$$P(I, D, G, S, L) = P(I)P(D)P(G \mid I, D)P(S \mid I)P(L \mid G)$$



Conditional Probability Distributions (CPDs)

- Proper CPD:

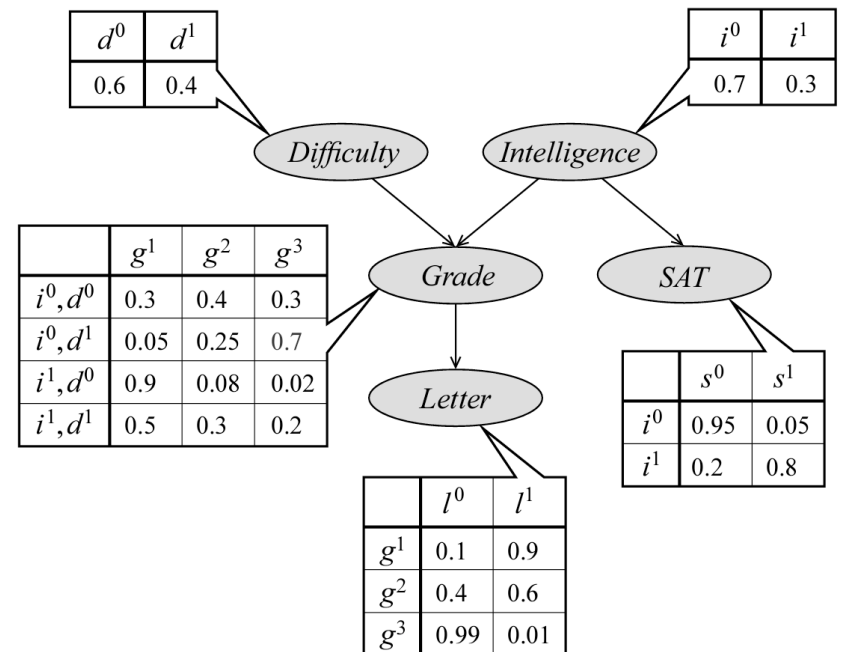
$$\sum_{x \in \text{Val}(X)} P(X = x \mid \mathbf{Parents}(X)) = 1$$

(for continuous case, integral)

- Everything breaks down into distributions over one variable given some others.
 - Factorization of high dimensional joint distribution $\rightarrow$ products of lower dimension CPDs (or factors)
 - Likelihood and Bayes rule can be used for making inference
- Let's examine CPDs in more details
 - Representation
 - Additional regularization

CPD representation

- So far, we've seen table representations
 - How many parameters?
 - Where will they come from?
 - Alternatives?
- Tabular CPD representation is not practical when we have many parents
 - # parameters grows exponentially with # parents
 - Medical application
- Tabular CPD ignore structure within the CPD



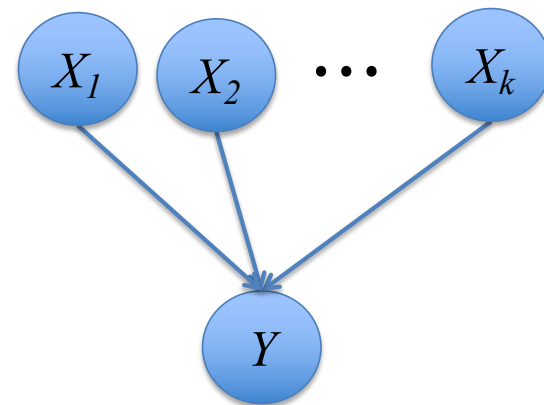
If we know more about the form of a CPD, we may be able to infer more properties of P.

Factorization and CPDs for casual influence

$$P(\mathbf{X}) = \prod_{i=1}^n P(X_i \mid \mathbf{Parents}(X_i))$$

Conditional Probability Distributions

How many parameters we need
for the causal influence for
binary random variables?



How can we represent CPDs?

- CPD $P(X \mid Y_1, \dots, Y_k)$ specifies distribution over X for each assignment $Y_1, \dots, Y_k$
- It does not have to use each value of X and Y_i explicitly
- CPDs can be seen as functions that given Y_i s and X , return the P
 - Can use any function to specify a factor $\phi(X, Y_1, \dots, Y_k)$ such that $\sum_x \phi(x, y_1, \dots, y_k) = 1$ for all $Y_1, \dots, Y_k$

Many Models can be used to represent CPDs

- Deterministic CPDs
- Tree-structured CPDs
- Logistic CPDs & generalizations
- Noisy OR / AND
- Linear Gaussians & generalizations

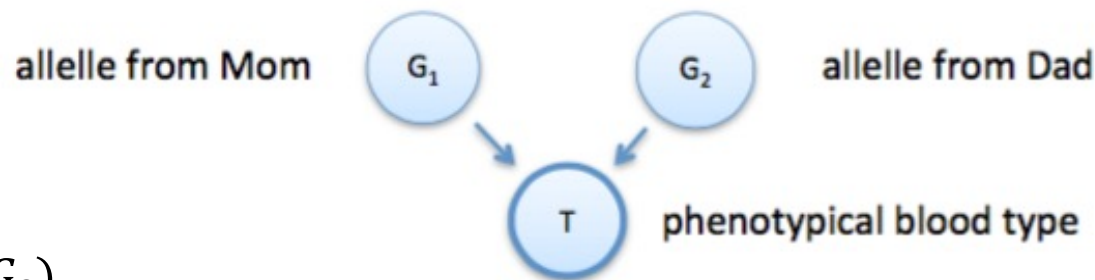
Deterministic CPDs

- Normally, CPDs in BN are stochastic
 - For each configuration of parent variables, the child variable has a distribution over possible outcomes.
- A deterministic CPD is a special case that the child's value is determined with certainty (probability = 1) by the values of its parents.
 - Instead of assigning probabilities, it specifies a deterministic function mapping parent values $\rightarrow$ child value.

Deterministic CPDs, example

A variable's value is a deterministic function of its parents' values.

Example:

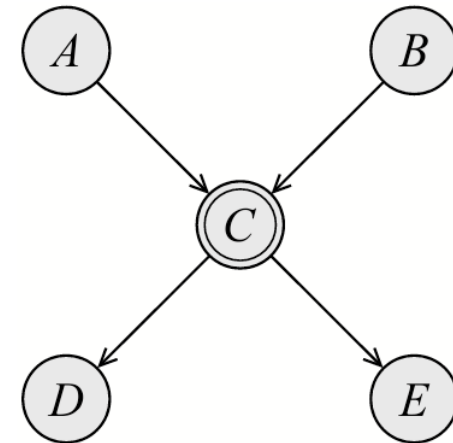


$$P(T|G_1, G_2)$$

$$T = \begin{cases} ab & \text{if } G_1 \text{ or } G_2 \text{ is } a \text{ and the other is } b \\ a & \text{if at least one of } G_1 \text{ or } G_2 \text{ is equal to } a \text{ and the other is either } a \text{ or } o \\ b & \text{if at least one of } G_1 \text{ or } G_2 \text{ is equal to } b \text{ and the other is either } b \text{ or } o \\ o & \text{if } G_1 = o \text{ and } G_2 = o \end{cases}$$

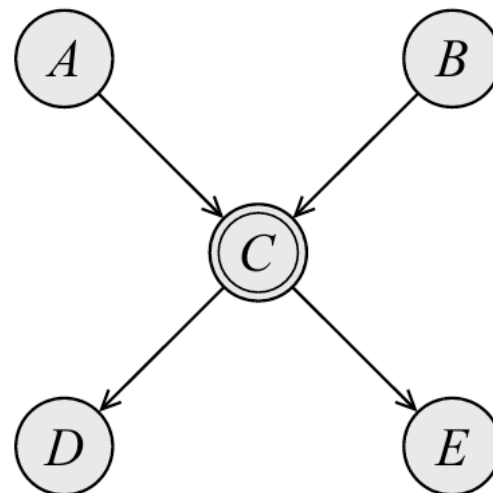
Deterministic CPDs, example

- $D \perp B | C, D \perp E | C$
local Markov assumption
 $(X_i \perp \text{NonDescendants}(X_i) | \text{Parents}(X_i))$
- $D \perp B | \{A, C\}$
 - Can we derive such independence assertions?
 - What if C is an OR function?
 - What about $D \perp E | A$?



Context-Specific Independence

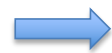
- Suppose $C = A \text{ OR } B$
- We are given $A=1$
 - This implies that $C = 1$
- $P(D|B,A=1) = P(D|A=1)$
 - B and D are Independent
- This does not work if $A=0$

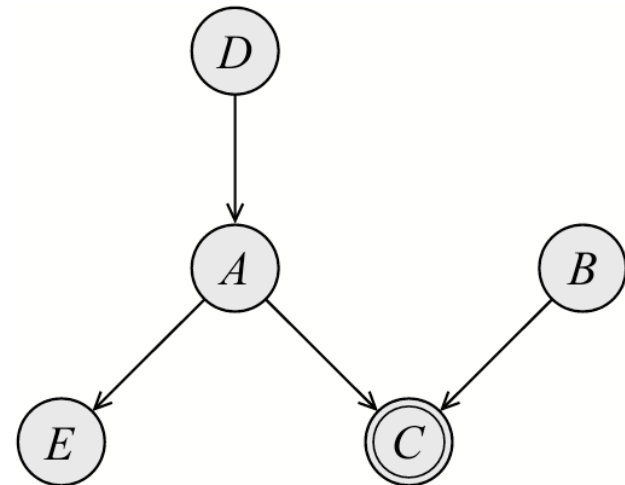


- Deterministic functions can imply a type of independence that holds only for particular values of some variables.
 - e.g. $(C \perp_c B \mid A=1)$, $(A \perp_c B \mid C=0)$,
- Deterministic CPDs gives a structure in CPDs

Another Example

- What can we say about $D \perp E | \{B, C\}$ in general?
- Suppose $C = A \text{ XOR } B$
- In the case of XOR,
 B and C fully determine A

 $D \perp E | \{B, C\}$



Context-Specific Independence

- Let X , Y , and Z be disjoint sets of variables; $\mathbf{C}$ is a set of variables that could overlap
- Let $c \in \text{Val}(\mathbf{C})$
- X and Y are contextually independent given Z and the context c denote $(X \perp_c Y \mid Z, c)$ if

$$P(\mathbf{X} \mid \mathbf{Y}, \mathbf{Z}, c) = P(\mathbf{X} \mid \mathbf{Z}, c) \text{ whenever } P(\mathbf{Y}, \mathbf{Z}, c) > 0.$$

Tree Structure CPDs

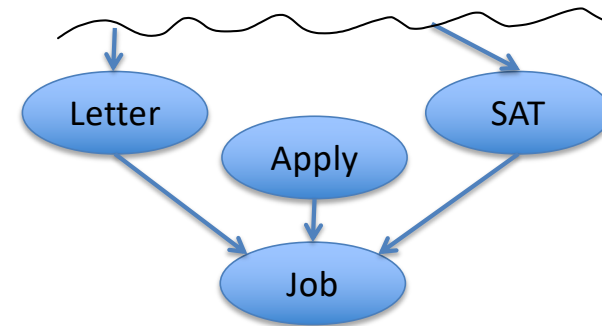
- We often have CPDs where several different distribution of $P(X|par(X))$ are identical

– Example

a) Among the 8 values of the parents of J , the four that have $A = 0$ have identical distribution over the variable J

b) If SAT score is high, Letter is not important $\rightarrow$

$$P(J | a^1, s^1, l^1) = P(J | a^1, s^1, l^0)$$

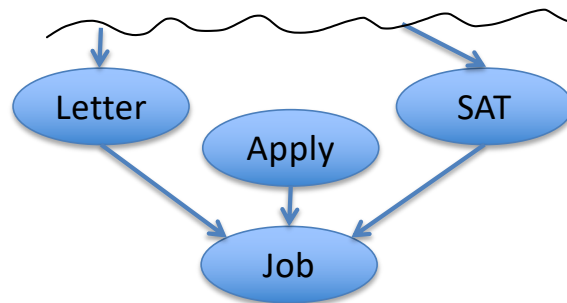


$$P(J | A, S, L)$$

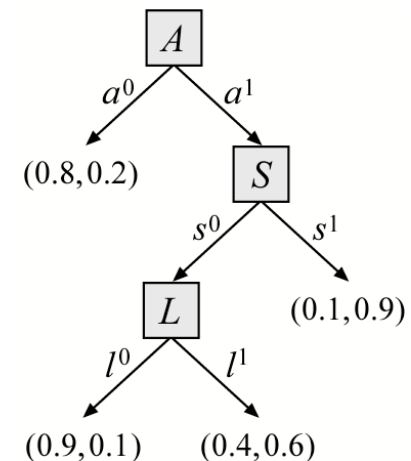
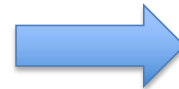
Use tree to capture regularities and common elements in a CPD

Use a tree to represent $P(X \mid \text{Parents}(X))$

- Represent the conditional distribution of a variable as a decision tree:
 - Internal nodes of the tree = parent variables (with tests like $X = x_1$).
 - Branches = different values of the parent variable, defining a context
 - Leaves = probability distributions over the child variable.
- Instead of specifying probabilities for every parent configuration, we group together cases that behave similarly.

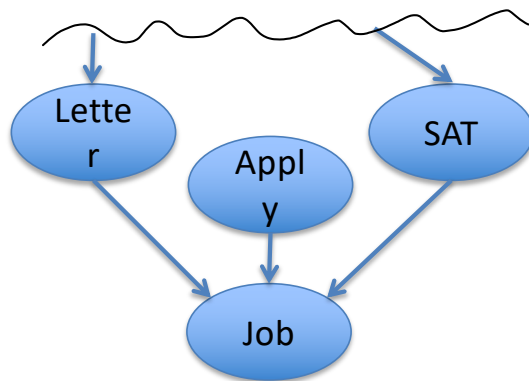


$$P(J \mid A, S, L)$$

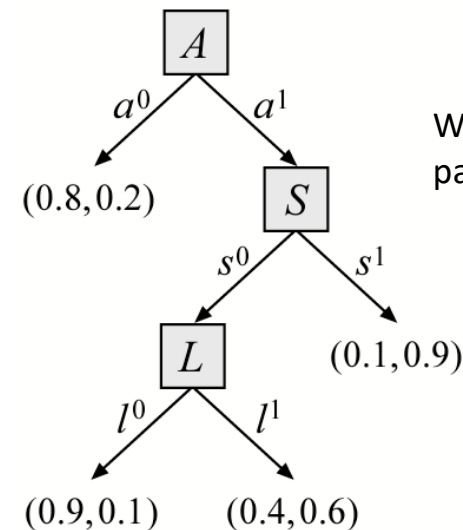
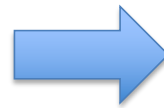


Why tree structured CPD?

- Compact representation: avoids exponential blowup.
- Interpretable: highlights which parents matter and in what order.
- Learnable: decision-tree learning algorithms can be adapted to learn tree-structured CPDs from data.



$$P(J \mid A, S, L)$$



We only need 4 parameters

Tree structured CPD, example

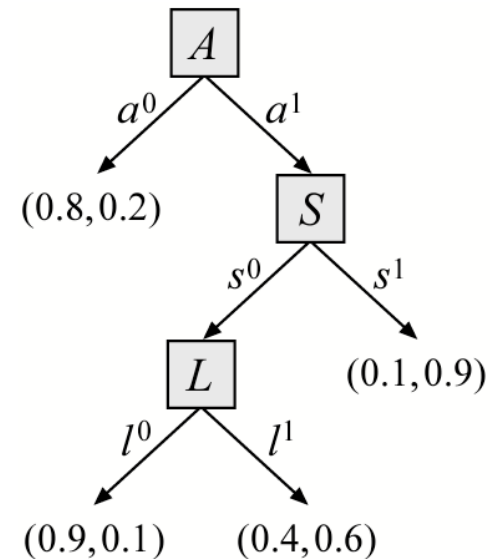
Which context-specific independence are implied by the structure of this CPD?

a) $(J \perp_c L \mid a^1, s^1)$

b) $(J \perp_c L \mid a^1)$

c) $(J \perp_c L, S \mid a^0)$

d) $(J \perp_c L \mid s^1, A)$

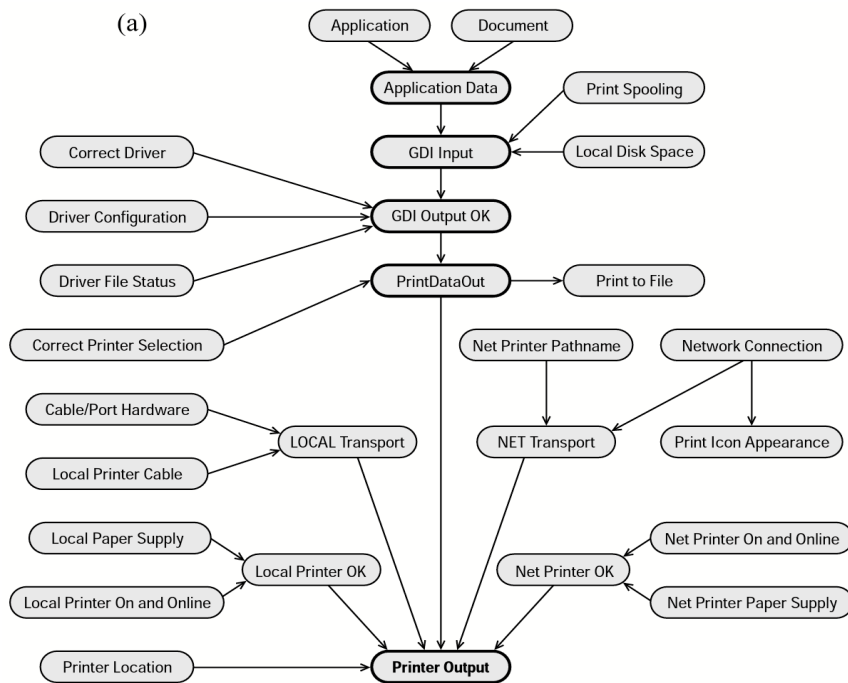


Tree CPD

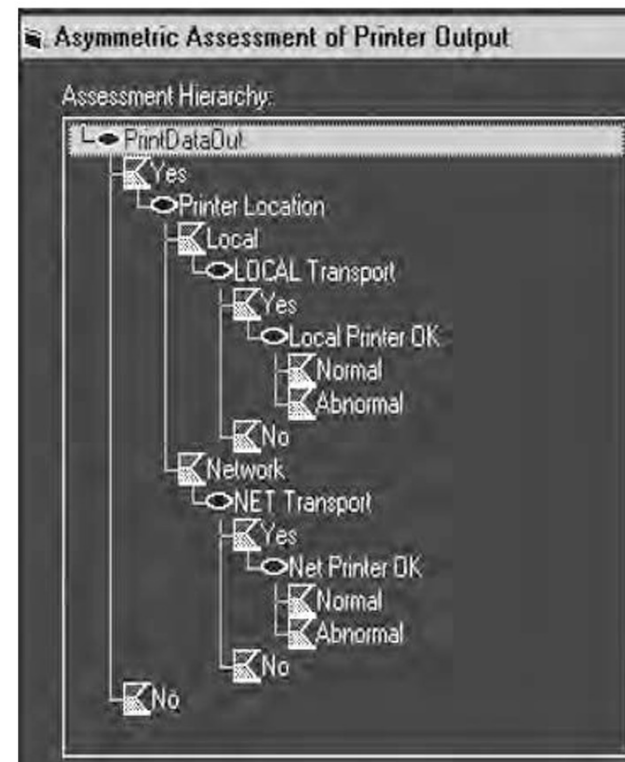
- Compact CPD representation that captures context-specific dependencies
- Relevant in multiple applications:
 - Hardware configuration variables
 - Medical settings
 - Dependence on agent's action
 - Perceptual ambiguity

Example: Microsoft Troubleshooting

Printer troubleshooting



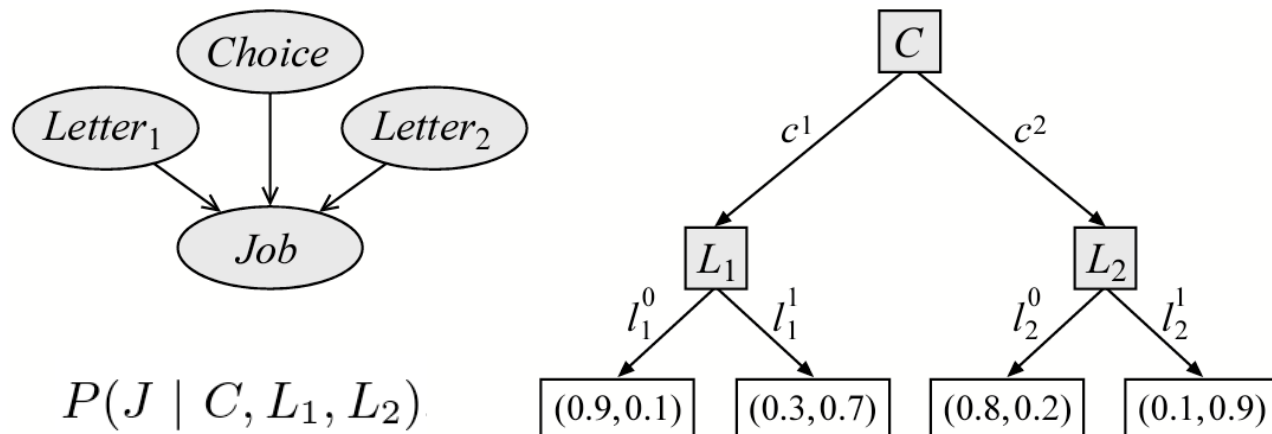
(b)



Number of parameters are reduced from 145 to 55

CPD representation: Multiplexer CPD

- A multiplexer CPD (also called a switching CPD) is a compact way to define the conditional probability of a variable when only one of its parents is relevant at a time, depending on the value of a “switch” (or selector) parent.

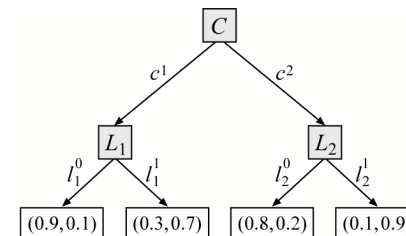
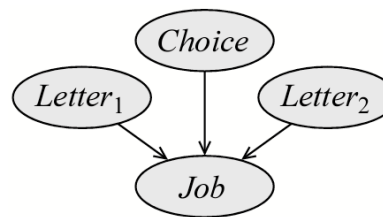


Depending on which choice, the dependence will only be on one of the letter

Multiplexer CPD

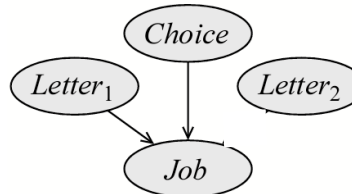
In general, we have the below dependency, even with the v-structure

$$(L_1 \perp L_2 \mid J, C)$$

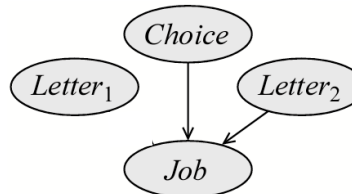


Context specific cases

$$(L_1 \perp_c L_2 \mid J, c_1)$$



$$(L_1 \perp_c L_2 \mid J, c_2)$$



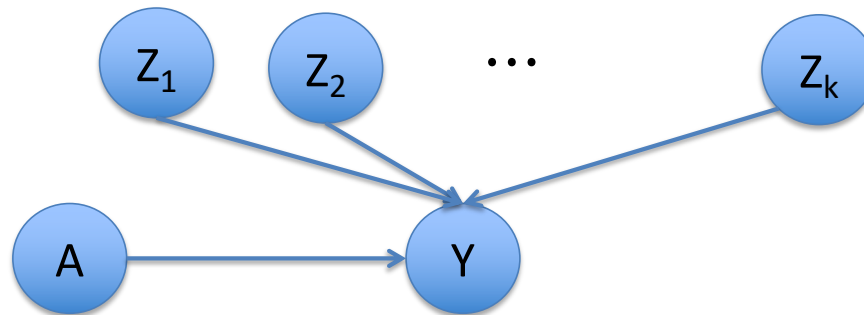
The active trail will disappear

Why Multiplexer CPD

- Compactness: avoids building a huge CPT over all parents.
- Interpretability: makes explicit that only one parent matters at a time.
- Applications: used in models of mixtures, switching systems, decision processes, and hierarchical Bayesian networks.
 - useful in domains where an outcome directly mirrors one parent chosen by context (e.g., routing, role selection, switching between sensors).

Multiplexer CPD

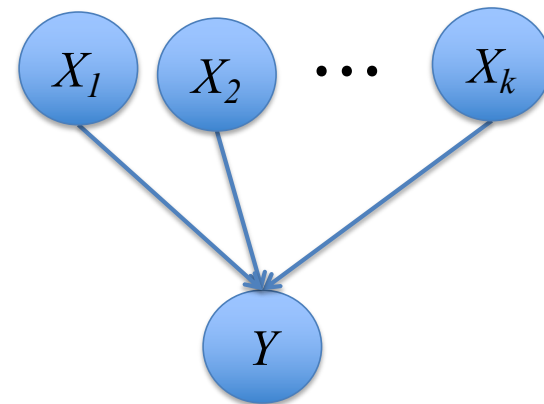
- A CPD $P(Y \mid A, Z_1, \dots, Z_k)$ is a multiplexer CPD if $\text{val}(A) = \{1, \dots, k\}$, and $P(Y \mid a, Z_1, \dots, Z_k) = \mathbf{I}\{Y = Z_a\}$
 - a is the value of A
 - A is called the selector



The value of the multiplexer variable is a copy of the value of one of its parents

Independence of causal influence

- Distribution of Y depends on some set of causes $X_1, \dots, X_k$
- Each X_i influence Y independently
 - Noisy OR model
 - Generalized linear model



Noisy OR Model, Example

- Two independent causal mechanisms by X_1, X_2

- Key assumptions

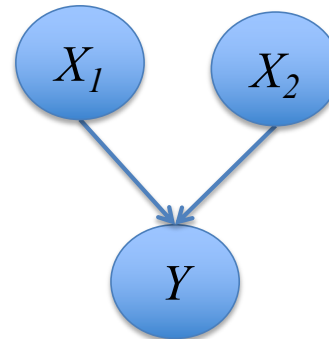
- “OR”: $Y = y^1$ cannot happen unless one of X_1, X_2 occurs

- “Noisy”: each causal mechanism is noisy.

- Independence of the causal mechanisms by X_1, X_2

$$P(Y = y^0 | X_1 = x_1^1, X_2 = x_2^1) = P(Y = y^0 | X_1 = x_1^0, X_2 = x_2^1) \times P(Y = y^0 | X_1 = x_1^1, X_2 = x_2^0)$$

X_1	X_2	Y	
		y^0	y^1
x_1^0	x_2^0	1	0
x_1^0	x_2^1	0.2	0.8
x_1^1	x_2^0	0.1	0.9
x_1^1	x_2^1	0.02	0.98



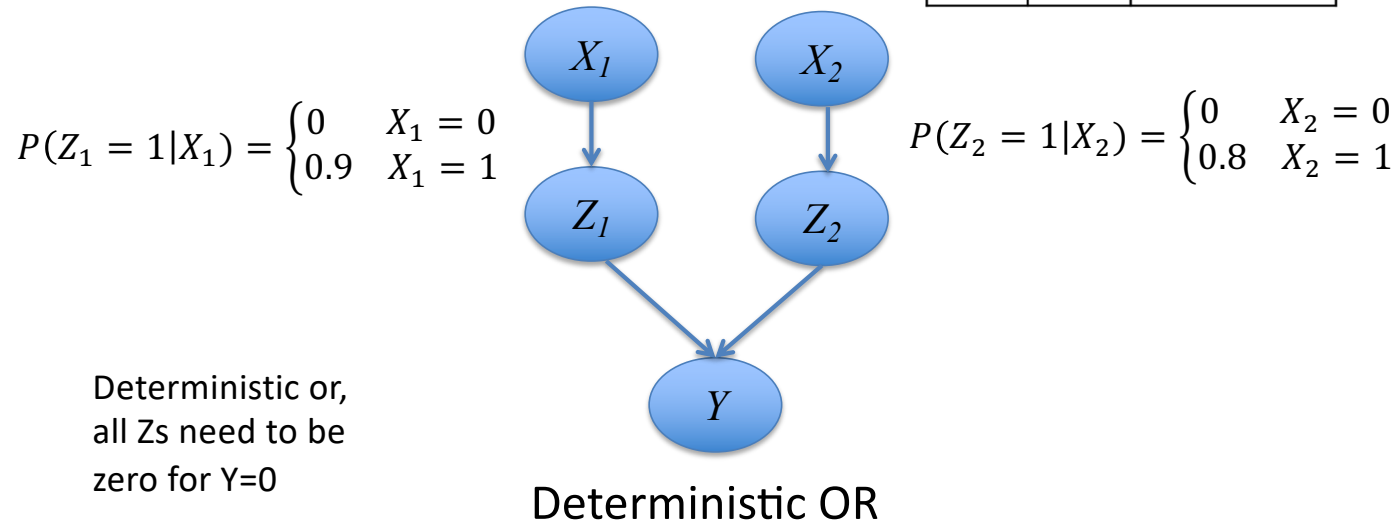
Noisy OR Model, Example

- Elaborate representation

Noise parameter $\lambda_1 = 0.9$

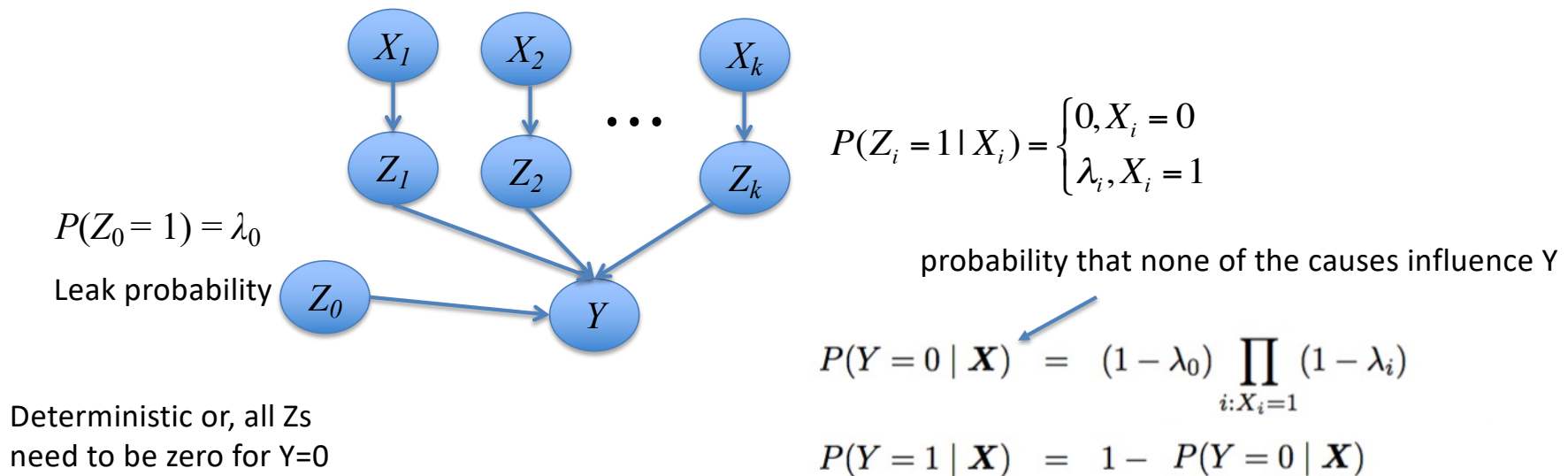
Noise parameter $\lambda_2 = 0.8$

$\mathbf{x_1}$	$\mathbf{x_2}$	$\mathbf{y}$	
		y^0	y^1
x_1^0	x_2^0	1	0
x_1^0	x_2^1	0.2	0.8
x_1^1	x_2^0	0.1	0.9
x_1^1	x_2^1	0.02	0.98



Noisy OR Model, generalized

- Conditional Bayesian network when multiple independent causes can trigger an effect
- Y and X_i s are binary random variables



Noisy OR Model

Y is a binary random variable with k binary parents, $x_1, \dots, x_k$, $P(Y \mid x_1, \dots, x_k)$ is a noisy OR if there are noisy parameters λ_i such that:

- λ_i is the noise parameter for X_i

$$P(Y = 0 \mid \mathbf{X}) = (1 - \lambda_0) \prod_{i: X_i=1} (1 - \lambda_i)$$

$$P(Y = 1 \mid \mathbf{X}) = 1 - \left[(1 - \lambda_0) \prod_{i: X_i=1} (1 - \lambda_i) \right]$$

- Or equivalently written (where $x^1 = 1$ and $x^0 = 0$)

$$P(y^0 \mid x_1, \dots, x_k) = (1 - \lambda_0) \prod_{i=1}^k (1 - \lambda_i)^{x_i}$$

Why Noisy OR model?

- Compact representation of CPDs, greatly reduces number of parameters.
 - Full table for n causes $\rightarrow 2^n$ entries.
 - Noisy-OR $\rightarrow$ only n parameters (one per cause, λ_i).
- Intuitive for causal reasoning (e.g., multiple diseases causing one symptom).
- Widely used in BNs for medical diagnosis, fault detection, etc.

Generalized linear model

- Satisfy Independence of Causal Influence

$$P(Y \mid X_1, \dots, X_k)$$

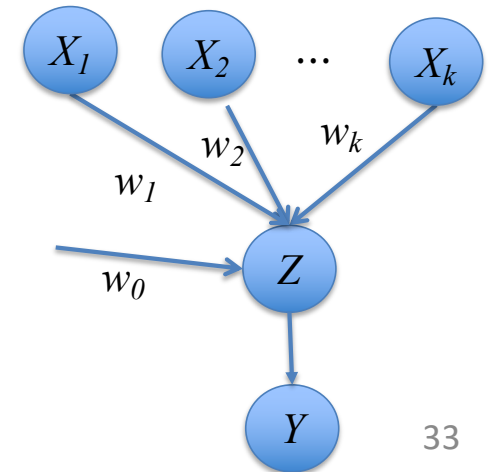
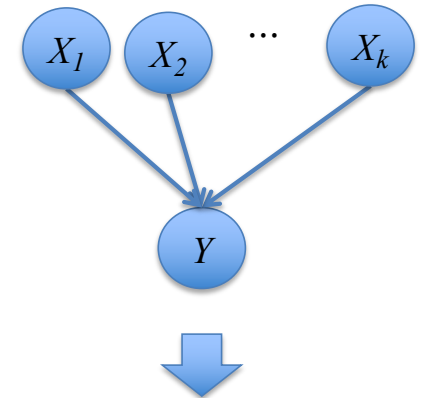
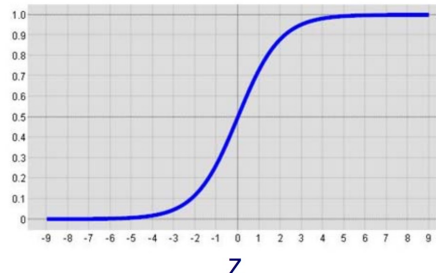
- Effects of X_i 's on $Z = f(X_1, \dots, X_k)$ is a linear function

$$f(X_1, \dots, X_k) = w_0 + \sum_{i=1}^k w_i X_i,$$

- The $CPD(Z \mid X_1, \dots, X_k)$ is deterministic

- Sigmoid or logit function to show how much $P(Y)$ depends on $f(X_1, \dots, X_k)$

$$\text{sigmoid}(z) = \frac{e^z}{1 + e^z}.$$



Generalized linear model

$$f(X_1, \dots, X_k) = w_0 + \sum_{i=1}^k w_i X_i,$$

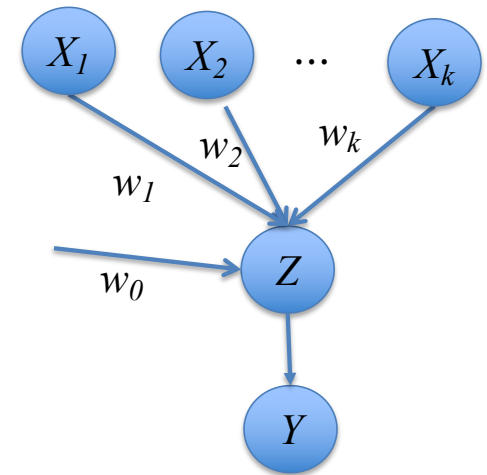
$$\text{sigmoid}(z) = \frac{e^z}{1 + e^z}.$$

- Sigmoid CPD

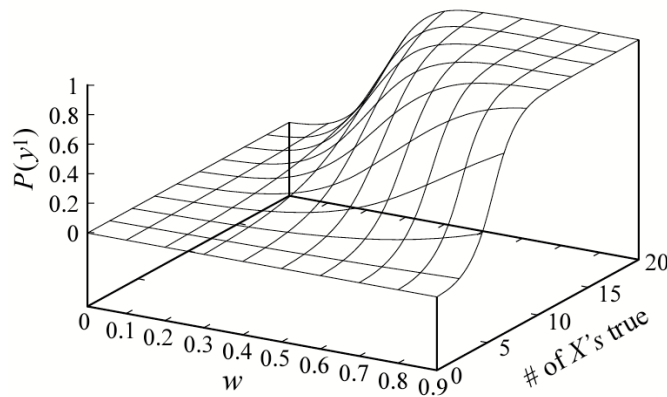
$$P(y^1 \mid X_1, \dots, X_k) = \text{sigmoid}\left(w_0 + \sum_{i=1}^k w_i X_i\right)$$

Odd ratio

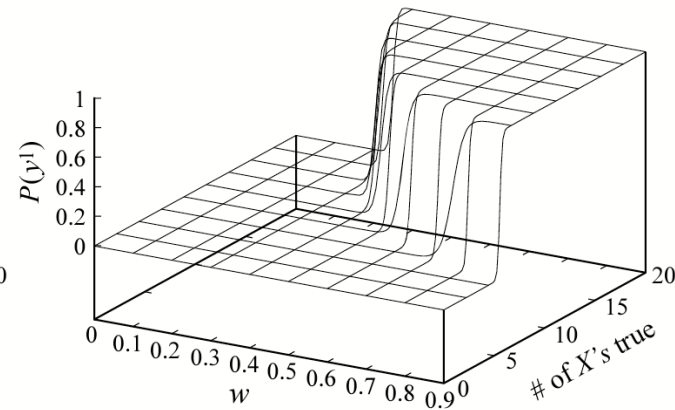
$$O(X) = \frac{P(Y^1|X)}{P(Y^0|X)} = \frac{\frac{e^Z}{1 + e^Z}}{1 - \frac{e^Z}{1 + e^Z}} = e^Z$$



Behavior of Sigmoid CPD



$w_0 = -5$



multiply w and w_0 by 10

More X true & larger w s $\rightarrow$ stronger $P(Y^1)$

Larger w $\rightarrow$ steeper transition

Continuous Random Variables

- One solution: discretize
 - Often requires too many value states
 - Loses domain structure
- Other solutions: use continuous function for $P(X | \text{Pa}(X))$
 - Can combine continuous and discrete variables, resulting in hybrid networks
 - Inference and learning may become more difficult

Continuous Random Variables

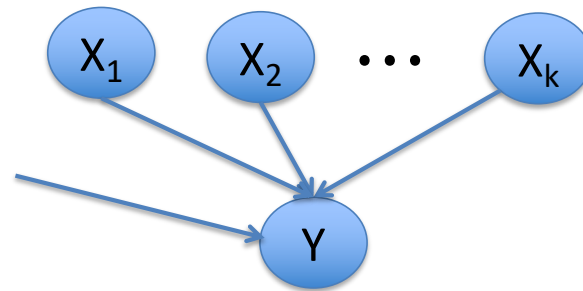
- Continuous child and parents

- Gaussian distribution is a CPD:

$$P(Y \mid \text{Mean} = \mu, \text{Variance} = \sigma^2) = \text{Normal}(\mu, \sigma^2)$$

- Linear Gaussian CPD: $\text{Normal}(\text{linear}(\mathbf{x}), \sigma^2)$
 - Y is a linear function of the variables $\mathbf{x}$, with Gaussian noise that has variance σ^2
- Linear Gaussian models are equivalent to multivariate Gaussian density functions

Linear Gaussian distribution



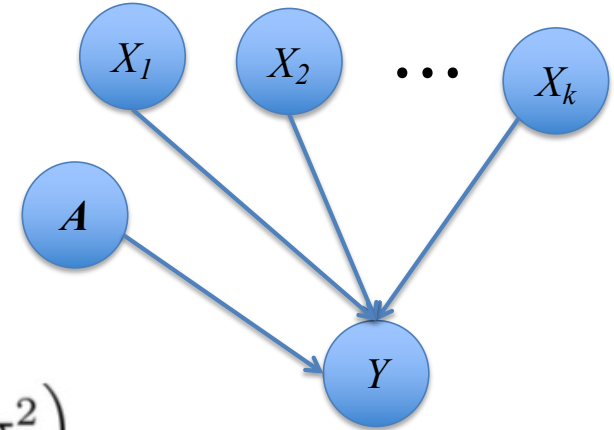
$$p(Y \mid x_1, \dots, x_k) = \mathcal{N}(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k; \sigma^2)$$

Vector representation

$$p(Y \mid \mathbf{x}) = \mathcal{N}(\beta_0 + \boldsymbol{\beta}^T \mathbf{x}; \sigma^2)$$

Conditional Linear Gaussian

- Continuous child with discrete and continuous parents
 - Y continuous variable
 - $\mathbf{X} = \{X_1, \dots, X_k\}$ continuous parents
 - $\mathbf{A} = \{a_1, \dots, a_m\}$ discrete parents
 - Set of coefficient w_{ai} and variance



$$Y \sim \mathcal{N} \left(w_{a0} + \sum w_{ai} X_i; \sigma_a^2 \right)$$

- Linear Gaussian whose parameters depend on A